

Harshita Arun

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PROFILE

Software Development Engineer (SDE-2) with 3+ years in payments and fintech at Blackbuck and Paytm. Built the BBPS and AutoPay (e-mandate) systems from scratch, covering settlement, reconciliation, retries and idempotency. Java, Spring Boot, Kafka, Redis, Elasticsearch, AWS, Kubernetes.

SKILLS

Programming: Java, Python, C/C++ | **Backend:** Spring Boot, Hibernate, REST, Microservices, Kafka, Redis
Data: MySQL, Elasticsearch, MongoDB, NoSql | **Cloud/DevOps:** AWS, Kubernetes, Git, Jenkins, Docker, CI/CD

WORK EXPERIENCE

Software Development Engineer 2 | Blackbuck *July 2026 - Present*

Software Development Engineer | Blackbuck *Mar 2025 - June 2026*

Payments | FASTag

- **Built the BBPS payment service from scratch** (HLD/LLD through production rollout) as **event-driven microservices** on Java, Spring Boot, Kafka, Redis, AWS and Kubernetes with **PayU integration**. It enables **third-party FASTag recharges across ~40 billers** with NPCI-compliant settlement at **~96% success**. It's built so other bill categories can be added later.
- **Architected and built the AutoPay (e-mandate) system from scratch**, with **Juspay integration** for mandate creation, tokenisation, and recurring payments. Designed the end-to-end transaction lifecycle and payment workflows, handling settlement, reconciliation and management. It handles **~3k new mandates and ~6k debit presentations a day at ~97% success**.
- **Owned non-functional work across the payment services:** parallelised a long-running cron job, cutting its runtime from 13+ hours to 3 hours (**~77%**), reduced latency from 2 minutes to under 10 seconds (**~82%**) **by fixing slow queries, resolved high-CPU alerts**, and added monitoring and alerting to improve overall system health.

Software Development Engineer | Paytm *July 2022 - Mar 2024*

Tech-Merchant team

- Built and maintained features on the merchant onboarding and management platform, handling over **0.5 million requests** daily.
- Built **GIS-based** agent assignment using a polygon territory management system.
- Optimised backend performance by migrating data retrieval workflows from relational databases to Elasticsearch, reducing response **times by 35%**.
- Worked on complex production issues, analysed system bottlenecks, implemented performance improvements, and contributed to continuous improvement initiatives focusing on scalability, maintainability and system reliability.

Software Engineer Intern (Paytm) *Jan 2022 - Jul 2022*

- As a back-end intern, worked actively in resolving software defects, ensuring the reliability of system components and compiled and maintained operational excellence reports, providing key insights into system efficiency

AI-Assisted Engineering

- **Co-developed an AI-assisted engineering harness using Claude Code**, standardizing the team's development workflow from PRD completeness and codebase analysis through implementation planning, approval gates, code generation, JUnit test creation, and release/ticket validation, with step-level execution logs for traceability.
- **Built end-to-end observability for FASTag purchase and recharge flows**, instrumenting each transaction step with state and failure annotations to pinpoint production failures; **developed a Claude Code debugging skill** that analyzes these logs and corresponding requests to identify the failed step and provide first-level root-cause diagnosis for support and engineering teams.

EDUCATION

Harcourt Butler Technical University (HBTU), UP 2018 - 2022 B.Tech.(Computer Science & Engineering)

Coursework: DSA, DBMS, OS, Computer Networks, C, C++, Algorithm analysis, Software Engineering

ACHIEVEMENTS/ CERTIFICATIONS

Stand Out Performer Award (Blackbuck). [link](#)

Rising Star Certificate (Paytm) [link](#) | 300+ coding problems solved | Technical Content Writer, Sahu Technologies

PROJECTS

• MiniRedis | Redis-inspired In-memory Key-Value Store([Project link](#))

Built a Redis-inspired in-memory key-value store in Java using TCP networking, data structures, LRU caching, TTL-based expiration, and concurrent client handling, supporting String, List, Set, and Hash data types, atomic INCR/DECR operations, configurable memory limits, and O(1) key lookup and eviction.

• Twitter Clone: Scalable Social Media Platform([Project link](#))

Developed a user friendly Twitter clone. It allows users to register, login, post tweets, retweet and view profile.